Phinma University of Iloilo
COLLEGE OF INFORMATION TECHNOLOGY EDUCATION
Rizal, Iloilo City, Philippines

CHAPTER IV

RESULTS AND DISCUSSION

4.1 Implementation

This chapter presents the results of the development and implementation of the Gonzaga Dental Clinic Management Automation System (GDCMAS). The system was developed using Next.js 16, React 19, TypeScript, Tailwind CSS 4, and Supabase as the backend database and authentication provider.

The implementation focused on digitizing and automating the clinic's core business processes, including patient registration, appointment scheduling, treatment recording, consent management, and inter-branch operations.

4.2 Development Tools and Technologies

The following tools and technologies were utilized in the development of GDCMAS:

Category	Tools/Technologies
Frontend Framework	Next.js 16, React 19, TypeScript
Styling	Tailwind CSS 4
Backend/Database	Supabase (PostgreSQL, Auth, RLS)
Validation	Zod
Development Environment	Visual Studio Code
Version Control	Git and GitHub

4.3 Hardware and Software Requirements

4.3.1 Hardware Requirements

Component	Minimum Requirement
Processor	Intel Core i5 or equivalent
RAM	8 GB or higher
Storage	256 GB or higher
Internet Connection	Stable broadband connection

4.3.2 Software Requirements

Software	Requirement
Operating System	Windows 10/11, Linux, or macOS
Web Browser	Google Chrome, Mozilla Firefox, or Microsoft Edge
Node.js	Version 18.18.0 or higher
Database	PostgreSQL 15+ (via Supabase)

4.4 System Features and Modules

• Patient Management Module

• Appointment Scheduling Module

• Treatment Records Module with FDI Tooth Chart

• Consent Management Module

• Public Booking Portal

• Staff Management Module

• Branch Management

• Audit Logging Module

4.5 Security Implementation

GDCMAS implements multi-layered security: Row Level Security on all 9 tables, three-layer authorization (Middleware $\rightarrow$ Server $\rightarrow$ RLS), role-based access control, branch-level isolation, audit logging, and database constraints preventing invalid operations.

4.6 Results Analysis

The system successfully addresses all identified problems. Centralized records eliminate fragmentation, automated scheduling prevents conflicts, role-based access ensures security, and audit logging provides accountability across both branches.

4.7 System Evaluation

The implementation meets all proposed objectives. Core modules are functional, security measures are enforced at the database level, and business processes are successfully digitized.

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CHAPTER V

CONCLUSION AND RECOMMENDATIONS

5.1 Introduction

This chapter presents the summary, findings, conclusions, and recommendations based on the development and implementation of GDCMAS.

5.2 Summary of the Study

The study aimed to develop and implement GDCMAS to automate Gonzaga Dental Clinic's business processes. Problems in the paper-based system were identified and addressed through a secure, web-based solution using modern technologies.

5.3 Findings

1. Paper-based system caused fragmentation and inefficiency.

2. Manual booking led to conflicts and duplicate records.

3. No backup or audit trail existed.

4. GDCMAS successfully centralized records with role-based access.

5. Multi-layered security and RLS protect sensitive data.

6. Database constraints ensure data integrity.

7. Audit logging ensures accountability.

5.4 Conclusion

GDCMAS effectively addresses all identified problems. It improves efficiency, accessibility, security, accountability, and patient care quality. The system provides enterprise-grade security while enabling seamless inter-branch operations, establishing a solid foundation for digital transformation.

5.5 Recommendations

1. Provide staff training on system usage.

2. Maintain regular backups via Supabase.

3. Deploy with proper domain and SSL.

4. Conduct periodic security reviews.

5. Consider future enhancements (notifications, reports, inventory).

6. Establish clear user management policies.

7. Perform regular updates and maintenance.